VocBee Production Readiness, Security, Code Quality ve UI/UX Denetim Raporu

Denetim tarihi: 3 Mayis 2026

Incelenen arsiv: vocbee-master.zip

Proje tipi: Expo / React Native mobil uygulamasi; Firebase Authentication + Firestore; RevenueCat; Google AdMob; Expo Notifications; Groq tabanli grammar proxy; 21 dilli i18n.

1. Yonetici ozeti

Bu kod tabani store kimlikleri, production EAS profili, App Store / Play Store URL leri ve temel production servislerini iceriyor. Buna ragmen denetim sonucum: su haliyle production-ready degil. Bunun nedeni tek bir problem degil; release gate, servis guvenligi, veri tutarliligi ve QA zincirinde birden fazla bloklayici var.

En kritik noktalar:

1. Release gate kirmizi: TypeScript, Jest, Firebase Functions build/lint ve npm audit basarisiz.
2. Android RevenueCat production key eksik: Android production build monetization akisi patlayabilir.
3. Grammar proxy herkese acik ve zayif korunuyor: maliyet, abuse, prompt injection ve hata sizintisi riski var.
4. Firestore rules arsivde yok: client dogrudan user dokumanlarini yaziyor; server-side access isolation kanitlanamiyor.
5. Sync katmaninda data loss riski var: favori sync hatasi yutuluyor; cloud restore local unsynced datayi ezebilir.
6. Account deletion akisi atomik degil: recent-login hatasinda Firestore silinip Auth kullanicisi kalabilir.
7. UI/i18n iyi bir temel iceriyor ama production polish eksik: 21 dil yapisal olarak tamam, fakat kalite, hardcoded metinler, RTL reload ve responsive testleri tamamlanmamis.

2. Kapsam ve metodoloji

Yapilan denetimler:

- Arsiv acildi ve proje mimarisi incelendi.
- Root ve functions bagimlilikleri kuruldu.
- npx tsc --noEmit, Jest testleri, Firebase Functions build/lint calistirildi.
- Root ve functions icin production dependency audit incelendi.
- Servis katmani, auth, sync, storage, monetization, ad, grammar proxy, i18n ve UI componentleri statik olarak incelendi.
- Locale dosyalari otomatik olarak key eksigi, bos deger, placeholder uyumsuzlugu ve English ile ayni kalan metinler acisindan kontrol edildi.

Sinirlar:

- Fiziksel cihaz/emulator ile tum ekranlar render edilmedi. UI/UX yorumu statik kod, testler ve responsive pattern denetimine dayanir.
- Firestore security rules, App Check ayarlari, RevenueCat dashboard, AdMob consent dashboard ve store console ayarlari ZIP icinde olmadigi icin dogrulanamadi.
- Bu rapor hukuki/uyumluluk danismanligi degildir; privacy/KVKK/GDPR metinleri icin hukuk gozden gecirmesi gerekir.

3. Otomatik kontrol sonuclari

Kontrol	Sonuc	Detay
Root dependency install	PASS	npm ci --ignore-scripts --no-audit --no-fund --progress=false tamamlandi.
TypeScript	FAIL	npx tsc --noEmit exit 2. Jest globals type disi; functions/src root tsconfig e dahil; tsconfig include/types ayrimi hatali.
Unit tests	FAIL	21 suite icinden 4 fail; 167 test icinden 12 fail. storage, streak, ErrorBoundary, OfflineNotice.
Functions build	FAIL	functions npm run build: unused onRequest/logger.
Functions lint	FAIL	unused import warning ve object-curly-spacing error.
Root npm audit prod	FAIL	{'info': 0, 'low': 0, 'moderate': 20, 'high': 7, 'critical': 1, 'total': 28}
Functions npm audit prod	FAIL	{'info': 0, 'low': 2, 'moderate': 11, 'high': 3, 'critical': 1, 'total': 17}
Locale structural check	PASS with warnings	21 dil, 343 key; missing/extra/empty/placeholder mismatch yok; bazi dillerde English ile ayni kalan keyler var.

4. Production-ready karari

Karar: Hayir, bu haliyle production release onerilmez.

Minimum release kosullari:

- TypeScript ve Jest temiz olmalı.
- Firebase Functions ya temiz build/lint ile deploy edilmeli ya da projeden/predeploy den kaldirilmali.
- Android RevenueCat key production EAS profile a eklenmeli.
- Grammar proxy server-side guvenlik katmani tamamlanmali.
- Firestore rules ve rule testleri eklenmeli.
- Critical/high npm audit bulgulari giderilmeli veya kabul edilebilir, belgelenmis, mitigation li hale getirilmeli.
- Account deletion ve sync conflict flowlari duzeltilmeli.

5. Mimari ve servis envanteri

Alan	Dosyalar	Gorev	Denetim yorumu
App Shell	App.tsx	Provider sirasi: SafeAreaProvider, ThemeProvider, AlertProvider, AuthProvider, MonetizationProvider, ErrorBoundary, OfflineNotice, AppContent.	Yapi mantikli; startup side effect ve loglar production gate ile kontrol edilmeli.
Authentication	context/AuthContext.tsx; services/nativeGoogleSignIn.ts	Firebase email/password, Google, Apple, profile cache, account deletion, sign out.	Deletion sirasi, local profile cleanup ve localized error handling iyilestirilmeli.
Firebase/Firestore Sync	services/SyncService.ts	Progress, favorites, profile, preferences, streak, notification settings sync.	Conflict resolution, failure propagation, security rules olmadan production riskli.
Local Storage	services/storage.ts; storageKeys.ts	AsyncStorage uzerinden data, progress, prefs, cache, sync flag.	PII minimizasyonu, profile_ cleanup, crypto random device id, SecureStore threat model.
Learning Progress	services/progressService.ts	Learned words, current level, reset, restore.	setCurrentLevel/resetProgress sync flag davranisi gozden gecirilmeli.
Favorites	services/favoritesService.ts	Local favorite list + Firebase sync.	sync error yutma, Firestore data validation ve duplicate cleanup onemli.
Streak	services/streakService.ts	Daily streak state, restore/clear, timezone-ish day logic.	Time injection ve restore semantics testleri duzeltilmeli.
Vocabulary	services/vocabularyService.ts; assets/vocabulary/*.txt	Bundled vocabulary JSON txt data, generation/cache.	Data schema validation ve localization quality gate eklenmeli.
Grammar AI	services/groqService.ts; api/grammarProxy.js; proxy/grammar-worker.ts	Client cache + proxy path + dev direct fallback.	Server side auth, durable rate limit, input/output validation, error sanitization gerekli.
Monetization	services/revenueCatService.ts; adMobService.ts;	Ad-free entitlement, paywall/customer center, ads.	Android RevenueCat key missing; consent/ad privacy tests

	context/MonetizationContext.tsx		gerekli.
Notifications	services/notificationService.ts; App.tsx	Permission gate, schedule/refill/process notifications.	Permission copy/localization, stale schedules and background sync tests gerekli.
i18n/RTL	services/i18n.ts; locales/*.ts; utils/rtl.ts	21 language resources, fallback en, RTL support.	Completeness iyi; quality/hardcoded text/RTL reload gate eksik.

6. Kritik bulgular tablosu

Seviye	Baslik	Bulgu	Kanıt	Production-ready duzeltme
CRITICAL	Release gate gecmiyor	TypeScript, Jest, Firebase Functions build/lint ve npm audit kontrolleri basarisiz.	tsconfig.json; jest-root.log; functions-build.log; functions-lint.log; npm-audit*.json	CI/CD pipeline zorunlu hale getirilmeli; kirmizi test/tsc/lint/audit duzelmeden release engellenmeli.
CRITICAL	Android RevenueCat production anahtari yok	eas.json production env sadece iOS RevenueCat anahtari iceriyor; Android tarafinda production build bos anahtar ile RevenueCat configure edemeyebilir.	eas.json:25-31; services/revenueCatService.ts:28-49	EXPO_PUBLIC_REVENUECAT_ANDROID_API_KEY veya ortak EXPO_PUBLIC_REVENUECAT_API_KEY production profile a eklenmeli; startup health check ve store smoke testi yapilmali.
HIGH	Grammar proxy herkese acik ve zayif korunuyor	CORS *; auth yok; kalici rate limit yok; input type/length/allowlist yok; hata mesajlari client a siziyor; Groq maliyet ve abuse riski var.	api/grammarProxy.js:1-20,53-97; proxy/grammar-worker.ts:19-31,35-66,68-139	Firebase Auth token/App Check veya imzali istek; durable rate limit; schema validation; length cap; outputLanguage allowlist; timeout; generic error response.
HIGH	Firestore security rules yok	ZIP icinde firestore.rules veya storage.rules bulunmadi. Client dogrudan users/{{uid}} dokumanlarini okuyor/yaziyor.	config/firebase.ts:8-15; services/SyncService.ts:108,148,204,229,348	Rules eklenip test edilmeli: request.auth.uid == uid, alan bazli validate, App Check enforcement, emulator rule tests.
HIGH	Sync hatasi favorilerde yutuluyor ve sync flag temizlenebiliyor	syncToFirestore favorites sync fonksiyonunu bekliyor; fakat syncFavoritesToFirebase kendi hatasini catch edip throw etmiyor. Sonra clearSyncFlag cagriliyor.	services/SyncService.ts:148-154; services/favoritesService.ts:219-235	syncFavoritesToFirebase boolean/throw donmeli; tum alt sync ler basarili olmadan LAST_SYNC/SYNC_FLAG temizlenmemeli.
HIGH	Account deletion sirasi veri tutarsizligi yaratabilir	Firestore dokumani once siliniyor, Auth kullanicisi sonra siliniyor. deleteUser requires-recent-login donerse Firestore silinmis, Auth kullanicisi kalmis olur.	context/AuthContext.tsx:622-635,645-650	Reauth gate i basta yap; once Auth deletion token uygunlugunu dogrula; sonra cloud data sil; tum adimlari idempotent server endpoint veya callable function ile yonet.
HIGH	Cloud restore local unsynced data yi ezebilir	fetchAndSaveUserData cloud verisini local e yazip clearSyncFlag cagiriyor. Yerel degisiklik daha yeni ise conflict resolution yok.	services/SyncService.ts:226-340,332	lastUpdated/version/deviceId alanlariyla merge stratejisi kur; restore oncesi local needsSync true ise cloud overwrite yapma veya conflict queue olustur.
HIGH	Dependency guvenlik borcu yuksek	Root prod audit: {'info': 0, 'low': 0, 'moderate': 20, 'high': 7, 'critical': 1, 'total': 28}. Functions prod audit: {'info': 0, 'low': 2, 'moderate': 11, 'high': 3, 'critical': 1, 'total': 17}.	npm-audit-root-prod.json; npm-audit-functions-prod.json	Expo/Firebase/axios zincirleri guncellenmeli; npm audit fix force dogrudan uygulanmadan Expo SDK uyumluluk matrisi ile kontrollu upgrade yapilmali.
MEDIUM	Firebase Functions scaffold deploy edilemiyor	functions/src/index.ts template; build unused imports ile, lint object spacing ile fail.	functions/src/index.ts:10-32; functions-build.log; functions-lint.log	Kullanilmiyorsa functions klasoru ve firebase predeploy kaldir; kullanilacaksa gercek callable/https fonksiyonu yaz, build/lint yesil yap.
MEDIUM	ErrorBoundary production da internal hata metnini gosteriyor	error.toString() kullanici ekranina basiliyor.	components/ErrorBoundary.tsx:71-75	Hata detayini sadece __DEV__ modda goster; production da genel mesaj ve crash reporting id goster.
MEDIUM	Android RECORD_AUDIO izni gereksiz gorunuyor	app.json RECORD_AUDIO tanimli; kodda aktif audio recording ozelligi gorulmedi.	app.json:82-85; privacypolicy.md:47-49	Kullanilmiyorsa izni kaldir; kullaniliyorsa feature, permission prompt ve privacy metni tam uyumlu olmali.
MEDIUM	In-app privacy metni guncele uymuyor	Locale privacy ozeti "veriler cihazda saklanir" diyor; uygulama Firestore cloud sync, AdMob, RevenueCat, Groq kullaniyor.	locales/en.ts:194-202; locales/tr.ts:privacy*; privacypolicy.md:79-85	In-app privacy modalini web privacy policy ile ayni veri kategorileri, bulut sync, ucuncu taraflar ve deletion haklariyla guncelle.
MEDIUM	Hidden OfflineNotice accessibility/test sorunu	Bagliyken component hala render ediliyor; opacity 0/pointerEvents none ama screen reader/test agacinda kalabilir.	components/OfflineNotice.tsx:34-59; components/__tests__/OfflineNotice.test.tsx	Connected durumda null don veya accessibilityElementsHidden/importantForAccessibility kullan; test beklentisini netlestir.
MEDIUM	RTL degisikligi reload gerektiriyor ama uygulanmiyor	applyRTLLayout true donuyor; i18n languageChanged sonucu reload tetiklemiyor.	services/i18n.ts:63-66; utils/rtl.ts:52-64	RTL/LTR degisiminde kullaniciya restart/reload akisi sun; UI elementlerinde start/end/marginLeft yerine logical spacing kullan.
MEDIUM	UI responsive riskleri	Bazi ekranlar module scope Dimensions.get kullaniyor; global display font cok yaygin; uzun ceviriler icin numberOfLines sinirli.	screens/HomeScreen.tsx:89; screens/SettingsScreen.tsx:55; screens/SetupScreens.tsx:51; utils/typography.ts; rg metrics	useWindowDimensions a gec; text scaling/large font testleri; tablet max-width stratejisi; uzun locale snapshotlari.
LOW	Hardcoded user-facing English messages	Google/Apple auth errorlari ve ExampleSentences accessibility label lar i18n disinda.	services/nativeGoogleSignIn.ts:103-127; context/AuthContext.tsx:489-490; screens/ExampleSentencesScreen.tsx:175,264,301,413,478	Tum hata ve accessibility metinleri locale dosyalarina tasinmali.

7. Servis bazli detayli analiz

7.1 Auth ve hesap yonetimi

Dosyalar: context/AuthContext.tsx, services/nativeGoogleSignIn.ts, services/googleAuthConfig.ts, config/firebase.ts.

Guclu yanlar:

- Email/password, Google ve Apple akislari var.
- Social sign-in sonrasi provider profilini Firestore/local profile ile merge etmeye calisiyor.
- Logout oncesi force sync dusunulmus.
- auth/requires-recent-login hatasi yakalaniyor.

Sorunlar:

- deleteAccount once Firestore dokumanini siliyor, sonra Firebase Auth kullanicisini siliyor (context/AuthContext.tsx:622-635). deleteUser recent-login isterse cloud dokumani silinmis ama auth kullanicisi kalmis olur. Bu hesap uyumlulugu ve destek acisindan problem yaratir.
- Apple sign-in genel hatasi hardcoded English (context/AuthContext.tsx:489-490). Google sign-in hata metinleri de i18n disinda (services/nativeGoogleSignIn.ts:103-127).
- Local profile cache profile_${uid} clearAllLocalData tarafindan temizlenmiyor; shared device senaryosunda eski profile kalma riski var (services/storageKeys.ts:47-49, services/storage.ts:118-151).

Duzeltme yaklasimi:

1. Delete flow icin once reauth kontrolu yap, sonra server-side idempotent deletion endpoint/callable function calistir.
2. Firestore delete ve Auth delete sirasini transactional/idempotent hale getir: deletionRequests/{uid} kaydi, Cloud Function cleanup, retry state.
3. Tum auth hata metinlerini locale keys e tasi.
4. clearAllLocalData icine profile_ prefix cleanup ekle.

7.2 Firebase ve Firestore guvenligi

Dosyalar: config/firebase.ts, services/SyncService.ts, services/favoritesService.ts.

Bulgu:

- Firebase client config client tarafinda bulunuyor. Bu tek basina secret degildir, fakat Firestore rules olmadan guvenlik saglamaz.
- ZIP icinde firestore.rules, storage.rules veya rule tests yok.
- Client users/{uid} dokumanini dogrudan set/get/delete ediyor.

Production-ready rules ornegi:

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    function signedIn() { return request.auth != null; }
    function isOwner(uid) { return signedIn() && request.auth.uid == uid; }

    match /users/{uid} {
      allow create: if isOwner(uid)
        && request.resource.data.email is string
        && request.resource.data.keys().hasOnly([
```

```
            'email','displayName','photoURL','createdAt','setupCompleted','preferences',
            'progress','streak','notificationSettings','lastActiveDeviceId'
          ]);
        allow read, update, delete: if isOwner(uid);
      }
    }
  }
```

Buna ek olarak @firebase/rules-unit-testing ile en az su testler olmalidir:

- Kullanici kendi dokumanini okuyabilir/yazabilir.
- Kullanici baska uid dokumanini okuyamaz/yazamaz/silemez.
- Bilinmeyen alanlar reddedilir.
- Progress/favorites/streak field tipleri hataliysa update reddedilir.

7.3 SyncService veri tutarliligi

Dosyalar: services/SyncService.ts, services/favoritesService.ts, services/progressService.ts, services/streakService.ts.

Guclu yanlar:

- Retry helper var.
- Progress, profile, favorites, preferences, notification settings ve streak tek sync akisi icinde dusunulmus.
- Device id ile last active device takip edilmeye calisilmis.

Sorunlar:

- shouldSync sadece 24 saat gecince sync ediyor (services/SyncService.ts:50-64). Kullanici uygulamayi silerse veya cihaz degistirirse 24 saat icinde kaybolan veriler olabilir. Logout force sync bunu tamamen cozmez.
- syncFavoritesToFirebase hata yakalayip throw etmiyor; syncToFirestore ise clearSyncFlag cagiriyor (services/SyncService.ts:150-154, services/favoritesService.ts:219-235). Bu data loss riskidir.
- Cloud restore local unsynced datayi ezebilir ve sync flag i temizler (services/SyncService.ts:226-340,332). Timestamp/version conflict strategy yok.
- setCurrentLevel sync flag isaretlemiyor (services/progressService.ts:140-146). resetProgress tek basina unsynced isaretlemiyor (services/progressService.ts:170-179).

Production-ready sync modeli:

- Her data domain e updatedAt, version, updatedByDeviceId ekle.
- Local mutation oldugunda sync queue olustur: pendingOps[] veya domain-level dirty flag.
- Sync 24 saat yerine debounce/throttle: foreground/background, network regain, mutation sonrasi 30-120 saniye.
- Cloud restore oncesi local dirty ise merge/conflict resolve et.
- Her alt sync fonksiyonu hata durumunda throw etsin; tum islem basarili olmadan sync flag temizlenmesin.

7.4 Grammar AI ve proxy guvenligi

Dosyalar: services/groqService.ts, api/grammarProxy.js, proxy/grammar-worker.ts.

Guclu yanlar:

- Production path proxy kullanacak sekilde tasarlanmis; Groq API key client ta tutulmak istenmemis.
- Client tarafinda timeout var (services/groqService.ts:164-166).
- AI response JSON parse ediliyor ve cache kullaniliyor.

Sorunlar:

- Vercel proxy ve Cloudflare Worker CORS * ile herkese acik (api/grammarProxy.js:2-5, proxy/grammar-worker.ts:35-41).
- Auth yok; uygulama disindan herkes endpoint i kullanabilir.
- Worker rate limit memory map; cold start ve multi-isolate durumunda guvenilir degil (proxy/grammar-worker.ts:19-31).
- Inputlar sadece presence ile kontrol ediliyor; type, length, allowed language yok (api/grammarProxy.js:16-20, proxy/grammar-worker.ts:59-66).
- outputLanguage prompt icine direkt giriyor; prompt injection riski var (api/grammarProxy.js:22-51, proxy/grammar-worker.ts:68-89).
- Hata mesajlari client a ham donuyor (api/grammarProxy.js:95-96, proxy/grammar-worker.ts:135-138).

Production-ready proxy iskeleti:

```
const ALLOWED_LANGUAGES = new Set(['Turkish','English','German','French','Spanish','Italian','Portuguese','Russian','Arabic','Chinese','Japanese','Korean']);
const MAX_SENTENCE_LEN = 280;

function validateBody(body: unknown) {
  if (!body || typeof body !== 'object') throw new HttpError(400, 'INVALID_BODY');
  const { sentence1, sentence2, outputLanguage = 'Turkish' } = body as any;
  if (typeof sentence1 !== 'string' || typeof sentence2 !== 'string') throw new HttpError(400, 'INVALID_SENTENCE');
  if (sentence1.trim().length < 1 || sentence1.length > MAX_SENTENCE_LEN) throw new HttpError(400, 'INVALID_SENTENCE');
  if (sentence2.trim().length < 1 || sentence2.length > MAX_SENTENCE_LEN) throw new HttpError(400, 'INVALID_SENTENCE');
  if (typeof outputLanguage !== 'string' || !ALLOWED_LANGUAGES.has(outputLanguage)) throw new HttpError(400, 'INVALID_LANGUAGE');
  return { sentence1: sentence1.trim(), sentence2: sentence2.trim(), outputLanguage };
}
```

Ek gereksinimler:

- Firebase ID token dogrulama veya App Check token.
- Durable Object/KV/Upstash tabanli kalici rate limit.
- AbortController timeout.
- Groq errorlarini sanitize et.
- Response schema validation: sentence1.structure|grammar|reason ve sentence2.* string ve length cap.

7.5 Monetization ve reklam

Dosyalar: services/revenueCatService.ts, context/MonetizationContext.tsx, services/adMobService.ts, app.json, eas.json.

Guclu yanlar:

- RevenueCat entitlement ad_free merkezi tanimlanmis.
- Production da test key kullanma guard i var (services/revenueCatService.ts:42-49).
- AdMob app ids ve ad unit ids ayrilmis.

Sorunlar:

- Android production RevenueCat key yok (eas.json:25-31, services/revenueCatService.ts:35-37). Bu P0 bug dir.
- TEST_REVENUECAT_API_KEY kaynak kodda duruyor. Guard var, fakat secrets/env yonetimi daha temiz olmali.
- Ad/ATT/UMP davranisi icin dashboard/config dogrulanamadi.

Duzeltme:

```
"production": {
  "env": {
    "EXPO_PUBLIC_GRAMMAR_PROXY_URL": "https://vocbee-proxy.vercel.app/api/grammarProxy",
```

```
      "EXPO_PUBLIC_REVENUECAT_IOS_API_KEY": "appl_...",
      "EXPO_PUBLIC_REVENUECAT_ANDROID_API_KEY": "goog_..."
    },
    "android": { "buildType": "app-bundle" }
  }
```

Smoke test listesi:

- Fresh install free user: ads gorunur, paywall acilir.
- Purchase monthly/yearly/lifetime: entitlement active, ads kalkar.
- Restore purchase: entitlement geri gelir.
- Logout/login same uid: RevenueCat logIn dogru calisir.
- Android production AAB internal testing: RevenueCat configure hatasi yok.

7.6 Storage ve local privacy

Dosyalar: services/storage.ts, services/storageKeys.ts, config/firebase.ts.

Sorunlar:

- AsyncStorage PII ve profil bilgileri icin kullaniliyor (services/storage.ts:90-107). Firebase Auth persistence de AsyncStorage (config/firebase.ts:21-26). Bu RN icin yaygin, fakat threat model dokumante edilmeli.
- Device id Date.now() + Math.random() ile uretiliyor (services/storage.ts:164-171). Security secret degil, ama collision/guessing acisindan expo-crypto daha iyi.
- clearAllLocalData profile prefix ve device id gibi bazi alanlari temizlemiyor (services/storage.ts:118-151). Logout icin device id kalabilir, ama delete account/shared device icin full purge secenegi olmalidir.

Duzeltme:

```
import * as Crypto from 'expo-crypto';

const profileKeys = allKeys.filter(key => key.startsWith(STORAGE_KEYS.PROFILE_PREFIX));
const reviewKeys = [STORAGE_KEYS.APP_REVIEW_PROMPT_STATE];

if (!deviceIdPreserve) keysToRemove.push(STORAGE_KEYS.DEVICE_ID);
keysToRemove.push(...profileKeys, ...reviewKeys);

const id = Crypto.randomUUID?.() ?? `device_${Date.now()}_${await Crypto.digestStringAsync(Crypto.CryptoDigestAlgorithm.SHA256, String(Math.random()))}`;
```

7.7 ErrorBoundary ve hata yonetimi

Dosya: components/ErrorBoundary.tsx.

Bulgu:

- error.toString() production ekraninda gosteriliyor (components/ErrorBoundary.tsx:71-75). Internal hata detayi kullaniciya ve ekran goruntulerine sizabilir.
- React Native SafeAreaView deprecated warning uretiyor; react-native-safe-area-context kullanilmali.
- Testler HOC export nedeniyle getDerivedStateFromError statigini bulamiyor.

Duzeltme:

```
{__DEV__ && this.state.error && (
  <View style={styles.errorBox}>
    <Text style={styles.errorText}>{this.state.error.toString()}</Text>
  </View>
```

```
  )}
```

Ayrica export { ErrorBoundaryInner } test icin ayrilabilir veya test HOC davranisina gore guncellenebilir.

8. UI/UX ve responsive tasarim denetimi

Pozitifler:

- Birden cok ekranda ScrollView, SafeAreaProvider ve responsive width hesaplari kullanilmis.
- iOS tablet destegi app config de acik (app.json:22-24).
- Dark/light theme yapisi var.
- Offline banner, error boundary, rating prompt gibi production UI durumlari dusunulmus.

Riskler:

- Dimensions.get('window') bircok dosyada module scope ta aliniyor (screens/HomeScreen.tsx:89, screens/SettingsScreen.tsx:55, screens/SetupScreens.tsx:51, components/HomeHeader.tsx:13). Orientation kilitli olsa da tablet split view, foldable, font scale ve runtime size change icin useWindowDimensions daha guvenilir.
- Global display font FONT_FAMILY cok yaygin kullaniliyor. Expanded display font uzun Almanca/Fransizca/Turkce/Hintce/Urdu metinlerde tasma riski yaratir.
- numberOfLines sinirli kullanilmis; Settings/Setup gibi uzun ceviri alanlarinda snapshot testi yok.
- OfflineNotice bagliyken de render agacinda duruyor; accessibility icin saklanmali veya null donmeli.
- RTL icin utility var, fakat forceRTL reload akisi yok. RTL layout degisiklikleri reload olmadan tam uygulanmaz (utils/rtl.ts:52-64).

Production UI test matrisi:

Boyut	Test
320-360px genis Android	Login, setup 7 adim, home card, examples modal, settings reset/delete modallari
iPhone SE	Keyboard/login/forgot password, paywall, modal overflow
Large iPhone/Android	Header alignment, card proportions, banner ad overlap
iPad/tablet	maxWidth, merkezleme, bosluklar, tablet orientation karari
Large text	system font scale 120%, 150%, 200%
RTL	Arabic/Farsi/Urdu onboarding, settings, chevrons, modal buttons
Dark mode	splash, error boundary, offline banner, paywall transition

9. i18n ve ceviri kalite denetimi

Otomatik sonuc:

- 21 dil dosyasi bulundu.
- English anahtar sayisi: 343.
- Tum dillerde missing/extra/empty key: 0.
- Placeholder mismatch: 0.
- Bazi dillerde English ile birebir ayni kalan metinler var; bir kismi marka/dil adi nedeniyle normal olabilir, bir kismi ceviri eksigi olabilir.

Dil	Key	Missing	Extra	Empty	English ile ayni	Ornekler
am	343	0	0	0	0	
ar	343	0	0	0	0	
de	343	0	0	0	8	languages.hindi, languages.swahili, languages.urdu, onboarding.start,

						settings.avatarOptionAccessibility, settings.min, settings.nameLabel, settings.support
es	343	0	0	0	7	auth.error, common.error, common.no, languages.hindi, languages.urdu, settings.avatarOptionAccessibility, settings.min
fa	343	0	0	0	0	
fr	343	0	0	0	6	example.structure, languages.hindi, languages.swahili, settings.min, settings.privacyContact, settings.support
hi	343	0	0	0	0	
id	343	0	0	0	5	auth.email, home.streak, languages.hindi, languages.swahili, languages.urdu
it	343	0	0	0	8	auth.email, auth.password, auth.passwordPlaceholder, common.no, languages.hindi, languages.swahili, languages.urdu, settings.min
ja	343	0	0	0	0	
ko	343	0	0	0	0	
pt	343	0	0	0	3	languages.hindi, languages.urdu, settings.min
ru	343	0	0	0	1	auth.email
sr	343	0	0	0	0	
sw	343	0	0	0	0	
th	343	0	0	0	0	
tr	343	0	0	0	1	settings.avatarOptionAccessibility
ur	343	0	0	0	0	
vi	343	0	0	0	1	auth.email
zh	343	0	0	0	0	

Ek bulgular:

- screens/ExampleSentencesScreen.tsx accessibility label lari hardcoded English: Play word pronunciation, Explain grammar with AI, Go back, Close modal, Retry loading explanation.
- Google/Apple auth hata metinleri i18n disinda.
- In-app privacy metni web privacy policy ile ayni seviyede guncel degil. Locale privacy ozeti cloud sync ve ucuncu taraflari yeterince aciklamiyor.
- RTL dilleri structural olarak listelenmis, ama reload ve layout mirror testleri eksik.

Production i18n gate:

1. i18n-lint script: missing/extra/empty/placeholder/same-as-English raporunu CI da calistir.
2. Hardcoded visible string taramasi: Text/Alert/accessibilityLabel icin i18n zorunlu.
3. Native speaker QA: en az tr, de, fr, es, ar, fa, ur, hi icin ekran bazli kontrol.
4. Pseudo-localization testi: uzun metinlerle UI tasma kontrolu.
5. RTL restart flow: languageChanged true donerse user a reload CTA.

10. Test ve CI duzeltme plani

10.1 TypeScript

Mevcut tsconfig.json root icinde tests ve functions i de kapsiyor; types sadece nativewind/types oldugu icin Jest global tipleri yok. Onerilen ayrim:

```
// tsconfig.app.json
{
  "extends": "./tsconfig.json",
  "include": ["App.tsx", "components/**/*.ts", "components/**/*.tsx", "screens/**/*.ts", "screens/**/*.tsx", "services/**/*.ts", "context/**/*.tsx", "config/**/*.ts",
"utils/**/*.ts", "types/**/*.d.ts"],
  "exclude": ["**/__tests__/**", "functions/**", "coverage/**", "node_modules/**"]
}
// tsconfig.test.json
{
  "extends": "./tsconfig.json",
  "compilerOptions": { "types": ["nativewind/types", "jest"] },
  "include": ["**/*.test.ts", "**/*.test.tsx", "jest.setup.js", "__mocks__/**/*.js"]
}
```

10.2 Test failures

- storage.test.ts: Deprecated saveLocalProgress/getLocalProgress exportlari silinmis. Testler yeni progressService e tasinmali veya compatibility wrapper geri eklenmeli.
- streakService.test.ts: Zaman bagimli restore semantigi bug/test uyumsuzlugu. now injection ile deterministik test yap.
- ErrorBoundary.test.tsx: HOC export nedeniyle static method yok. Inner class export et veya test HOC davranisini test et.
- OfflineNotice.test.tsx: Component bagliyken null donmuyor. Product karari: animasyon icin render kalacaksa accessibility hidden; test buna gore guncellenmeli.

10.3 CI pipeline onerisi

```
name: mobile-ci
on: [pull_request]
jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
        with:
          node-version: 22
          cache: npm
      - run: npm ci
      - run: npm run typecheck
      - run: npm test -- --runInBand
      - run: npm audit --omit=dev --audit-level=high
      - run: npm --prefix functions ci
      - run: npm --prefix functions run lint
      - run: npm --prefix functions run build
```

11. Oncelikli duzeltme yol haritasi

Oncelik	Alan	Aksiyon
P0 - Release blocker	CI gate	tsc/test/lint/build/audit fail durumlarini release i engelleyecek pipeline a bagla.
P0 - Revenue	Android RevenueCat	Production Android API key ekle; iOS/Android store sandbox purchase/restore smoke testleri yaz.
P0 - Security	Grammar proxy	Auth/App Check + Durable Object/KV rate limit + schema validation + generic errors.
P0 - Data isolation	Firestore rules	Rules ve emulator tests ekle, sadece users/{auth.uid} erisimi.
P1 - Data integrity	Sync	Failure propagation, conflict resolution, sync flag semantics, deletion order.
P1 - Privacy	Policy and permissions	RECORD_AUDIO kararini ver; in-app privacy copy guncelle; account deletion recent-login flow.
P2 - UI/i18n	Responsive and localization	useWindowDimensions, font fallback, RTL reload, hardcoded labels.
P2 - QA	Device matrix	Small Android, iPhone SE, large iPhone, iPad/tablet, RTL, large text, dark mode.

12. Release checklist

Build health

- ☐ npm ci clean install
- ☐ npx tsc --noEmit yesil
- ☐ npm test -- --runInBand yesil
- ☐ functions npm run lint/build yesil veya functions kaldirildi
- ☐ npm audit production icin kabul edilebilir seviyede ve documented

Security

- ☐ Firestore rules deployed and emulator-tested
- ☐ App Check enabled for Firestore/Functions where supported
- ☐ Grammar proxy auth/rate-limit/schema/timeout/generic-error implemented
- ☐ No server secrets in EXPO_PUBLIC or client bundle
- ☐ Dependency critical/high vulnerabilities fixed or formally accepted with mitigation

Data integrity

- ☐ Sync failure propagation tested
- ☐ Conflict resolution tested across two devices
- ☐ Logout/delete/reinstall flows tested
- ☐ Offline -> online queue tested
- ☐ Account deletion idempotent and reauth-safe

Monetization/ads

- ☐ RevenueCat iOS and Android keys set
- ☐ Entitlement id ad_free verified

- ☐ Purchase/restore/customer center tested
- ☐ AdMob app ids/ad unit ids verified
- ☐ UMP/ATT consent flows tested

UI/UX

- ☐ Small/large screen snapshots
- ☐ Tablet support decision documented
- ☐ Large text accessibility tested
- ☐ Dark/light mode tested
- ☐ RTL Arabic/Farsi/Urdu tested after app reload
- ☐ Offline banner accessibility tested

Localization/legal

- ☐ All hardcoded user strings extracted
- ☐ Same-as-English list reviewed by native speakers
- ☐ In-app privacy copy aligned with privacy policy
- ☐ Store permission descriptions aligned with actual features

13. Kanit log ozetleri

13.1 Jest final summary

```
      expect(received).toBeNull()

    Received: {"_fiber": {"_debugHookTypes": null, "_debugInfo": null, "_debugNeedsRemount": false, "_debugOwner": [FiberNode], "_debugStack": [Error: react-stack-top-frame],
  "_debugTask": null, "actualDuration": 0, "actualStartTime": 1777819905079, "alternate": null, "child": [FiberNode], "childLanes": 0, "deletions": null, "dependencies": null,
  "elementType": "Text", "flags": 0, "index": 0, "key": null, "lanes": 0, "memoizedProps": [Object], "memoizedState": null, "mode": 3, "pendingProps": [Object], "ref": null,
  "refCleanup": null, "return": [FiberNode], "selfBaseDuration": 0, "sibling": null, "stateNode": [Object], "subtreeFlags": 0, "tag": 5, "treeBaseDuration": 0, "type": "Text",
  "updateQueue": null}}

      111 |         });
      112 |
    > 113 |         expect(queryByText('offline.message')).toBeNull();
          |                                                ^
      114 |
      115 |         // Simulate disconnect
      116 |         mockGetNetworkStateAsync.mockResolvedValue({ isConnected: false });

      at Object.toBeNull (components/__tests__/OfflineNotice.test.tsx:113:48)
      at asyncGeneratorStep (node_modules/@babel/runtime/helpers/asyncToGenerator.js:3:17)
      at _next (node_modules/@babel/runtime/helpers/asyncToGenerator.js:17:9)

  ● OfflineNotice › handles undefined connection state
```

```
    expect(received).toBeNull()

    Received: {"_fiber": {"_debugHookTypes": null, "_debugInfo": null, "_debugNeedsRemount": false, "_debugOwner": [FiberNode], "_debugStack": [Error: react-stack-top-frame],
"_debugTask": null, "actualDuration": 0, "actualStartTime": 1777819905114, "alternate": null, "child": [FiberNode], "childLanes": 0, "deletions": null, "dependencies": null,
"elementType": "Text", "flags": 0, "index": 0, "key": null, "lanes": 0, "memoizedProps": [Object], "memoizedState": null, "mode": 3, "pendingProps": [Object], "ref": null,
"refCleanup": null, "return": [FiberNode], "selfBaseDuration": 0, "sibling": null, "stateNode": [Object], "subtreeFlags": 0, "tag": 5, "treeBaseDuration": 0, "type": "Text",
"updateQueue": null}}

      135 |
      136 |         // isConnected !== false, so should be treated as connected
    > 137 |         expect(queryByText('offline.message')).toBeNull();
          |                                                ^
      138 |     });
      139 | });
      140 |

      at Object.toBeNull (components/__tests__/OfflineNotice.test.tsx:137:48)
      at asyncGeneratorStep (node_modules/@babel/runtime/helpers/asyncToGenerator.js:3:17)
      at _next (node_modules/@babel/runtime/helpers/asyncToGenerator.js:17:9)


Test Suites: 4 failed, 17 passed, 21 total
Tests:       12 failed, 155 passed, 167 total
Snapshots:   0 total
Time:        27.993 s
Ran all test suites.
```

13.2 Functions build

```
> build
> tsc

src/index.ts(11,1): error TS6133: 'onRequest' is declared but its value is never read.
src/index.ts(12,1): error TS6133: 'logger' is declared but its value is never read.
```

13.3 Functions lint

```
> lint
> eslint --ext .js,.ts .


/mnt/data/vocbee_work/vocbee-master/functions/src/index.ts
  11:9   warning  'onRequest' is defined but never used  @typescript-eslint/no-unused-vars
  12:13  warning  'logger' is defined but never used     @typescript-eslint/no-unused-vars
  27:19  error    There should be no space after '{'     object-curly-spacing
  27:36  error    There should be no space before '}'    object-curly-spacing

✖ 4 problems (2 errors, 2 warnings)
  2 errors and 0 warnings potentially fixable with the `--fix` option.
```

14. Sonuc

VocBee mimari olarak dogru yonde: Expo, Firebase, RevenueCat, AdMob, i18n, notification ve error boundary gibi production uygulamalarinda gereken parcalar mevcut. Ancak production hazirligi sadece bu parcalarin varligi degildir. Release icin otomatik kalite kapilari yesil olmali, server-side guvenlik kanitlanmali, veri sync tutarliligi saglanmali, privacy metinleri gercek veri akisina uygun olmali ve UI tum cihaz/dil kombinasyonlarinda test edilmelidir.

Bu rapordaki P0 maddeleri tamamlanmadan release onerilmez.

VocBee Production-Ready Duzeltme Rehberi

Bu dokuman, denetim raporundaki bulgularin uygulanabilir duzeltme adimlarina indirgenmis halidir. Siralama, release blocker riskine gore yapildi.

P0. Release gate i yesile cevir

1. Package scripts ekle

package.json icine:

```
{
  "scripts": {
    "typecheck": "tsc -p tsconfig.app.json --noEmit",
    "typecheck:test": "tsc -p tsconfig.test.json --noEmit",
    "test:ci": "jest --runInBand",
    "audit:prod": "npm audit --omit=dev --audit-level=high",
    "functions:lint": "npm --prefix functions run lint",
    "functions:build": "npm --prefix functions run build"
  }
}
```

2. tsconfig ayrimi

- Root tsconfig.json sadece ortak compilerOptions tutsun.
- App icin tsconfig.app.json tests/functions exclude etsin.
- Test icin tsconfig.test.json Jest tiplerini eklesin.
- Functions kendi tsconfig i ile derlensin, root include a girmesin.

3. Jest fail fixleri

storage.test.ts

Secenek A: Testleri yeni progressService davranisina tasi.

Secenek B: Backward compatible wrapper ekle:

```
export const saveLocalProgress = async (progress: Partial<UserProgress>): Promise<void> => {
  const existing = await getLocalProgress();
  const next = { ...existing, ...progress, lastUpdated: new Date().toISOString() };
  await AsyncStorage.setItem(LOCAL_KEYS.USER_PROGRESS, JSON.stringify(next));
  await markAsUnsynced();
};

export const getLocalProgress = async (): Promise<UserProgress> => {
  try {
    const data = await AsyncStorage.getItem(LOCAL_KEYS.USER_PROGRESS);
```

```
    return data ? JSON.parse(data) : { currentLevel: 'A1', learnedWords: {}, lastUpdated: new Date().toISOString() };
  } catch {
    return { currentLevel: 'A1', learnedWords: {}, lastUpdated: new Date().toISOString() };
  }
};
```

ErrorBoundary.test.tsx

```
export class ErrorBoundaryInner extends Component<InnerProps, State> { ... }
export const ErrorBoundary = withThemeColors(ErrorBoundaryInner);
```

Test static method i ErrorBoundaryInner.getDerivedStateFromError uzerinden test etsin.

OfflineNotice.test.tsx

Product karari:

```
if (isConnected && !hasEverBeenOffline) return null;
```

veya:

```
<Animated.View
  accessibilityElementsHidden={isConnected}
  importantForAccessibility={isConnected ? 'no-hide-descendants' : 'auto'}
/>
```

streakService.test.ts

getTodayKey(now = new Date()) veya service factory ile zaman inject et. Fake timers ile restore testini deterministik yap.

P0. RevenueCat Android production key

eas.json:

```
"production": {
  "env": {
    "EXPO_PUBLIC_GRAMMAR_PROXY_URL": "https://vocbee-proxy.vercel.app/api/grammarProxy",
    "EXPO_PUBLIC_REVENUECAT_IOS_API_KEY": "appl_...",
    "EXPO_PUBLIC_REVENUECAT_ANDROID_API_KEY": "goog_..."
  },
  "android": { "buildType": "app-bundle" }
}
```

Startup guard:

```
if (!__DEV__ && Platform.OS === 'android' && !process.env.EXPO_PUBLIC_REVENUECAT_ANDROID_API_KEY) {
  throw new Error('Missing Android RevenueCat key');
}
```

P0. Grammar proxy hardening

Server-side kontrol listesi

- ☐ POST disi 405.
- ☐ `Content-Type: application/json` zorunlu.
- ☐ Body max 2 KB.
- ☐ sentence1/sentence2 string, trim, 1-280 char.
- ☐ outputLanguage allowlist.
- ☐ Firebase ID token veya App Check token dogrulama.
- ☐ Durable rate limit: uid + ip bazli.
- ☐ AbortController timeout 10-15 sn.
- ☐ Groq error sanitize.
- ☐ JSON schema output validate.
- ☐ Audit logs PII siz.

Cloudflare Worker shape

```
type GrammarResponse = {
  sentence1: { structure: string; grammar: string; reason: string };
  sentence2: { structure: string; grammar: string; reason: string };
};

function assertGrammarResponse(value: unknown): GrammarResponse {
  const v = value as any;
  for (const s of ['sentence1','sentence2']) {
    for (const k of ['structure','grammar','reason']) {
      if (typeof v?.[s]?.[k] !== 'string' || v[s][k].length > 1200) {
        throw new HttpError(502, 'BAD_AI_RESPONSE');
      }
    }
  }
  return v;
}
```

P0. Firestore rules ve tests

Dosya ekle: firestore.rules. CI da emulator testlerini calistir.

Test senaryolari:

- Kendi dokumanini okuma/yazma success.
- Baska uid dokumanina read/write denied.
- Auth olmadan denied.
- progress.learnedWords invalid type denied.
- preferences.level invalid denied.

- streak.currentStreak negative denied.

P1. Sync data integrity

favorites sync throw etsin

```
export const syncFavoritesToFirebase = async (): Promise<void> => {
  const user = auth.currentUser;
  if (!user) throw new Error('AUTH_REQUIRED');
  const favorites = await getFavorites();
  await setDoc(doc(db, 'users', user.uid), { favorites, favoritesUpdatedAt: serverTimestamp() }, { merge: true });
};
```

sync flag temizleme kuralı

```
const results = await Promise.allSettled([
  syncProgress(),
  syncFavoritesToFirebase(),
  syncStreak(),
]);
const failed = results.filter(r => r.status === 'rejected');
if (failed.length) throw new Error('PARTIAL_SYNC_FAILED');
await setLastSyncTime();
await clearSyncFlag();
```

Conflict strategy

- Her domain icin updatedAt ve version.
- Local dirty ise cloud restore yapma; once merge.
- Multi-device conflict te son yazan degil, domain merge: favorites union, progress union, streak max/best, preferences explicit user action.

P1. Account deletion

Onerilen akış:

1. Kullanici reauth yapar.
2. Server/callable deletion request olusturulur.
3. Cloud Function Firestore ve Storage cleanup yapar.
4. Auth user delete edilir.
5. Local full purge yapilir.

Client-only kalacaksa en azindan:

```
await reauthenticateUser();
await deleteUser(user);          // recent-login burada patlar; cloud silinmemis olur
await deleteDoc(userDocRef);     // sonra cloud cleanup
```

```
await clearAllLocalData({ fullPurge: true });
```

P1. Privacy ve izinler

- RECORD_AUDIO kullanilmiyorsa app.json dan kaldir.
- In-app privacy modalini web policy ile ayni bilgilerle guncelle: Firebase, Firestore, RevenueCat, AdMob, Groq, Expo, local storage, deletion, retention.
- Store permission descriptions ile gercek feature lar ayni olsun.

P2. UI/UX standardizasyon

- Module scope Dimensions.get yerine useWindowDimensions.
- Ana layoutlarda maxWidth ve tablet strategy.
- Display font sadece heading/brand; body metinlerde okunabilir font.
- Long translation snapshot: de/fr/tr/ar/ur/fa/hi.
- Large text accessibility: fontScale 1.3/1.6/2.0.
- RTL reload prompt.

P2. Localization

- accessibilityLabel ve error mesajlarini locale e tasi.
- Same-as-English raporunu native speaker ile temizle.
- Pseudo-localization script ekle.

Kabul kriteri

Aşağıdakiler yesil olmadan release yok:

```
npm ci
npm run typecheck
npm run typecheck:test
npm run test:ci
npm run audit:prod
npm --prefix functions ci
npm --prefix functions run lint
npm --prefix functions run build
```

VocBee Production Release Checklist

Build health

- ☐ npm ci clean install
- ☐ npx tsc --noEmit yesil
- ☐ npm test -- --runInBand yesil
- ☐ functions npm run lint/build yesil veya functions kaldirildi
- ☐ npm audit production icin kabul edilebilir seviyede ve documented

Security

- ☐ Firestore rules deployed and emulator-tested
- ☐ App Check enabled for Firestore/Functions where supported
- ☐ Grammar proxy auth/rate-limit/schema/timeout/generic-error implemented
- ☐ No server secrets in EXPO_PUBLIC or client bundle
- ☐ Dependency critical/high vulnerabilities fixed or formally accepted with mitigation

Data integrity

- ☐ Sync failure propagation tested
- ☐ Conflict resolution tested across two devices
- ☐ Logout/delete/reinstall flows tested
- ☐ Offline -> online queue tested
- ☐ Account deletion idempotent and reauth-safe

Monetization/ads

- ☐ RevenueCat iOS and Android keys set
- ☐ Entitlement id ad_free verified
- ☐ Purchase/restore/customer center tested
- ☐ AdMob app ids/ad unit ids verified
- ☐ UMP/ATT consent flows tested

UI/UX

- ☐ Small/large screen snapshots
- ☐ Tablet support decision documented
- ☐ Large text accessibility tested
- ☐ Dark/light mode tested
- ☐ RTL Arabic/Farsi/Urdu tested after app reload
- ☐ Offline banner accessibility tested

Localization/legal

- ☐ All hardcoded user strings extracted
- ☐ Same-as-English list reviewed by native speakers
- ☐ In-app privacy copy aligned with privacy policy
- ☐ Store permission descriptions aligned with actual features

Go/No-Go

- ☐ P0 bulgular kapatildi.
- ☐ P1 bulgular icin fix veya kabul edilmis risk dokumani var.
- ☐ Store internal test build iOS ve Android smoke testlerden gecti.
- ☐ Firebase/RevenueCat/AdMob/Groq dashboard ayarlari production ortaminda dogrulandi.